

Lecture 2

Kendra Burbank

Outline

This lecture covers several foundational concepts that connect real analysis to optimization:

1. Properties of Differentiable Functions

- Mean Value Theorem and its implications
- Monotonicity and derivative signs
- First and second-order optimality conditions
- Taylor's Theorem for function approximation

2. Lipschitz Functions

- Definition and properties
- Connection to bounded derivatives
- Error bounds for linear approximations

3. **Newton's Method**

- Formulation and geometric intuition
- Application to root-finding problems
- Quadratic convergence properties

4. **Analysis of Newton's Method**

- Understanding convergence behavior
- Error analysis using Taylor's Theorem

Properties of Differentiable Functions

The Mean Value Theorem is one of the most important results in calculus, and serves as a foundation for many optimization techniques.

MVT connects the average rate of change of a function over an interval with the instantaneous rate of change at some point within that interval.

Mean value theorem

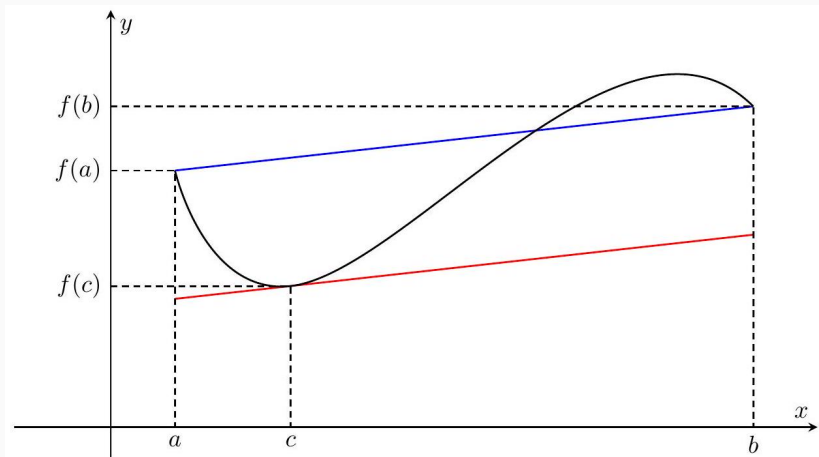


Figure 1: Mean value theorem: slope of tangent $f'(c) = [f(b) - f(a)]/(b - a)$ slope of secant.

Formally:

Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Corollary

If f is differentiable on (a, b) , then

- 1. $f'(x) \geq 0$ for all $x \in (a, b) \Rightarrow f$ is monotone increasing;*
- 2. $f'(x) = 0$ for all $x \in (a, b) \Rightarrow f$ is constant;*
- 3. $f'(x) \leq 0$ for all $x \in (a, b) \Rightarrow f$ is monotone decreasing.*

The converse of these statements are also true.

A function is said to be **monotone increasing** (or *nondecreasing*) if

$$x_1 < x_2 \quad \Rightarrow \quad f(x_1) \leq f(x_2),$$

and **monotone decreasing** (or *nonincreasing*) if

$$x_1 < x_2 \quad \Rightarrow \quad f(x_1) \geq f(x_2)$$

We say *strictly increasing* or *strictly decreasing* if the inequalities on the right hand sides are strict.

Proof of Corollary 2.1

Suppose for a continuous and differentiable function f we have $f'(x) \geq 0$ for all $x \in (a, b)$. We want to show that f is monotone increasing.

Choose any two points $x_1, x_2 \in (a, b)$ with $x_1 < x_2$.

Since f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) , by the mean value theorem, there exists c with $x_1 < c < x_2$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \Rightarrow f'(c)(x_2 - x_1) = f(x_2) - f(x_1).$$

Now we know by supposition that $f'(c) \geq 0$ and that $x_2 > x_1$, so the product $f'(c)(x_2 - x_1) \geq 0$.

Therefore, $f(x_2) - f(x_1) \geq 0$, i.e., $f(x_2) \geq f(x_1)$.

The other statements are proved similarly.

Prove the converse as an exercise and deduce similar statements for $f'(x) > 0$ and $f'(x) < 0$.

Checking for local optima

The next two results help us ascertain whether a local optimizer is a minimizer or maximizer when the first derivative is available in a neighborhood of the optimizer.

Theorem

Suppose f is continuous at x_0 and differentiable in a neighborhood of x_0 . Then

*If $f'(x) < 0$ in a left neighborhood of x_0 and $f'(x) > 0$ in a right neighborhood of x_0 , then x_0 is a **minimizer**;*

*If $f'(x) > 0$ in a left neighborhood of x_0 and $f'(x) < 0$ in a right neighborhood of x_0 , then x_0 is a **maximizer**.*

Note that we don't require differentiability at x_0 itself to apply this result. For example, we can deduce that $f(x) = |x|$ has a minimizer at $x = 0$ by applying Theorem 3.1.

Second order sufficient condition for local optimality

If the function is twice differentiable at a critical point, we can use the second derivative to check for local optimality.

If the second derivative is **positive** at a critical point, the function is a local **minimizer**; if the second derivative is **negative**, the function is a local **maximizer**. Formally:

Theorem

Let $f \in C^2(X)$ and x_0 be a local optimizer, so $f'(x_0) = 0$. Then

- *if $f''(x_0) > 0$, then x_0 is a **minimizer**;*
- *if $f''(x_0) < 0$, then x_0 is a **maximizer**.*

A common rookie mistake is to overestimate the usefulness of Theorem 3.2 and underestimate that of Theorem 3.1, as you will see in Homework 1.

Prove Theorems Theorem 3.1 and Theorem 3.2.

Taylor's theorem

Taylor's theorem is a generalization of the Mean Value Theorem. It allows us to approximate functions using polynomials of various degrees, which is crucial for many optimization algorithms.

The theorem says we can approximate a function with the first n terms of its Taylor polynomial, and then lets us describe the error in the approximation using a remainder term that we can specify using the higher order derivatives.

By looking closely at the remainder term, we'll be able to do things like bound the error in the approximation.

Formally,

Theorem

If f has continuous derivatives of order up to and including $n + 1$ in some neighborhood of x_0 , then

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x),$$

where the remainder term is

$$\begin{aligned} R_n(x) &= \frac{1}{n!} \int_{x_0}^x f^{(n+1)}(t)(x-t)^n dt && \text{(integral form)} \\ &= \frac{1}{n!} f^{(n+1)}(\xi) \frac{(x-x_0)^{n+1}}{n+1} && \text{(mean value form)} \end{aligned}$$

for some ξ between x_0 and x .

- The integral form is usually much more useful in optimization (partly because we don't have to know ξ !).
- The proof of Taylor's theorem is from the Fundamental Theorem of Calculus. We won't do it here.
- Preview: Next lecture, we'll use Taylor's theorems on functions whose first derivatives are Lipschitz continuous. These functions are more regular than C^1 functions, but not as regular as C^2 functions; we can call them $C^{1.5}$ functions.

Properties of Lipschitz functions

Remember, a function $f : X \rightarrow \mathbb{R}$ is said to be *Lipschitz* if there is some $\gamma < \infty$ such that

$$|f(x) - f(y)| \leq \gamma|x - y|$$

for all $x, y \in X$. We write $f \in \text{Lip}_\gamma(X)$.

For differentiable functions, having a bounded derivative is equivalent to being Lipschitz continuous. Intuitively, if the derivative is bounded, then the function can't change too rapidly between any two points. Formally:

Lemma

Let $f : X \subset \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. Then f' is bounded if and only if f is Lipschitz on X .

Proof of Lemma 5.1

First, we will show that if the derivative is bounded, then the function is Lipschitz.

Suppose that the derivative is bounded, i.e. there exists some γ such that $|f'(x)| \leq \gamma$ for all x .

Let $x_1, x_2 \in X$ be distinct.

By the Mean Value Theorem, there exists ξ between x_1 and x_2 such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(\xi)$$

Taking absolute values and re-arranging,

$$\frac{|f(x_2) - f(x_1)|}{|x_2 - x_1|} = |f'(\xi)| \leq \gamma \quad \Rightarrow \quad |f(x_2) - f(x_1)| \leq \gamma |x_2 - x_1|$$

Hence $f \in \text{Lip}_\gamma(X)$.

Now we will show that if the function is Lipschitz, then the derivative is bounded.

Suppose that $f \in \text{Lip}_\gamma(X)$. Then

$$|f(x) - f(x_0)| \leq \gamma |x - x_0|$$

for every x and x_0 in X .

Re-arranging,

$$\left| \frac{f(x) - f(x_0)}{x - x_0} \right| \leq \gamma$$

The inequality holds as $x \rightarrow x_0$, so

$$|f'(x)| = \left| \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \right| = \lim_{x \rightarrow x_0} \left| \frac{f(x) - f(x_0)}{x - x_0} \right| < \gamma$$

In optimization, the Lipschitz condition usually arises in the derivative, not the function.

That is, $f' \in \text{Lip}_\gamma(X)$.

This is a stronger condition than $f \in C^1(X)$ but weaker than $f \in C^2(X)$

Turns out to be the right level of regularity for optimization.

Assuming that X is compact, then

$$f \in C^2(X) \Rightarrow f' \in \text{Lip}_\gamma(X) \Rightarrow f \in C^1(X)$$

so the condition $f' \in \text{Lip}_\gamma(X)$ is a bit like saying that " $f \in C^{1.5}(X)$."

A bound on the error of the linear approximation

Many optimization algorithms use the linear approximation of a function. The better this approximation is, the better the algorithm will perform.

Here, we put a bound on the error of the linear approximation for functions that are C^1 and have a Lipschitz continuous derivative.

Suppose we approximate $f(x)$ using the linear approximation $f(x_0) + f'(x_0)(x - x_0)$. This is what you'd get if $f(x)$ was just a straight line.

How bad can this approximation be?

We approximate the derivative $f'(x)$ with the constant $f'(x_0)$.

The bigger the difference between $f'(x_0)$ and $f'(x)$, the worse the approximation.

The error in the approximation of $f(x)$ is the integral of the error in the derivative:
$$\int_{x_0}^x f'(z) - f'(x_0) dz.$$

Lipschitz continuity tells us that the most the derivative can change as we move from x_0 to x is $\gamma \times (x - x_0)$.

So the error in the linear approximation is at most the integral of this: $\frac{1}{2}\gamma(x - x_0)^2$.

Formally,

Lemma

Let $f : X \rightarrow \mathbb{R}$ be C^1 and $f' \in \text{Lip}_\gamma(X)$. Then for any $x_0, x \in X$,

$$|f(x) - f(x_0) - f'(x_0)(x - x_0)| \leq \frac{\gamma}{2}(x - x_0)^2.$$

Proof of Lemma 5.2

The formal proof uses a change of variables so that we can integrate over the interval $[0, 1]$ instead of $[x_0, x]$:

By the Fundamental Theorem of Calculus,

$$f(x) - f(x_0) = \int_{x_0}^x f'(z) dz$$

Next we will subtract $f'(x_0)(x - x_0)$ from both sides. (This lets us calculate the error in the linear approximation.)

Note that $f'(x_0)(x - x_0) = \int_{x_0}^x f'(x_0) dz$ (since $f'(x_0)$ is constant), so

$$f(x) - f(x_0) - f'(x_0)(x - x_0) = \int_{x_0}^x f'(z) dz - \int_{x_0}^x f'(x_0) dz = \int_{x_0}^x f'(z) - f'(x_0) dz$$

Applying the change of variables $z = x_0 + t(x - x_0)$ (and thus $dz = (x - x_0)dt$),

$$\int_{x_0}^x f'(z) - f'(x_0) dz = \int_0^1 (f'(x_0 + t(x - x_0)) - f'(x_0)) (x - x_0) dt.$$

Taking absolute values and applying the Lipschitz condition on f' ,

$$\begin{aligned} |f(x) - f(x_0) - f'(x_0)(x - x_0)| &= \left| \int_0^1 (f'(x_0 + t(x - x_0)) - f'(x_0)) (x - x_0) dt \right| \\ &\leq \int_0^1 |f'(x_0 + t(x - x_0)) - f'(x_0)| |x - x_0| dt \\ &\leq \int_0^1 \gamma |t(x - x_0)| |x - x_0| dt \\ &= \gamma |x - x_0|^2 \cdot \int_0^1 t dt \\ &= \frac{\gamma}{2} (x - x_0)^2. \end{aligned}$$

Note that this lemma is stronger than Taylor's Theorem since we do not require f to be twice differentiable. Why is this important? The lemma gives us a linear approximation of f :

$$f(y) = f(x) + f'(x)(y - x) + \text{error}$$

where for some constant C ,

$$| \text{error} | \leq \frac{C}{2} |y - x|^2 .$$

This means that we can use many optimization algorithms that are based on linear approximations even if we don't have f twice differentiable.

Newton's method

The first optimization algorithm we'll look at in detail is **Newton's method**.

Newton's method is known for its rapid convergence properties. It uses local quadratic approximations to “jump” directly toward the optimum, rather than taking small steps like gradient descent.

Goal: Find a root of a nonlinear equation $g(x) = 0$.

In the context of optimization, the function is usually some derivative, i.e., $g = f'$.

In the multivariate case, Newton's method is used to find roots of system of nonlinear equations:

$$\begin{cases} g_1(x_1, \dots, x_n) = 0 \\ g_2(x_1, \dots, x_n) = 0 \\ \vdots \\ g_n(x_1, \dots, x_n) = 0 \end{cases}$$

Intuitively, the steps of Newton's method are like this:

1. Start with an initial guess x_0
2. Find the tangent line to $g(x)$ at x_0
3. Find the root of the tangent line. (This is using the linear approximation of $g(x)$ at x_0 .)
4. Use the root as the next guess x_1
5. Repeat until convergence

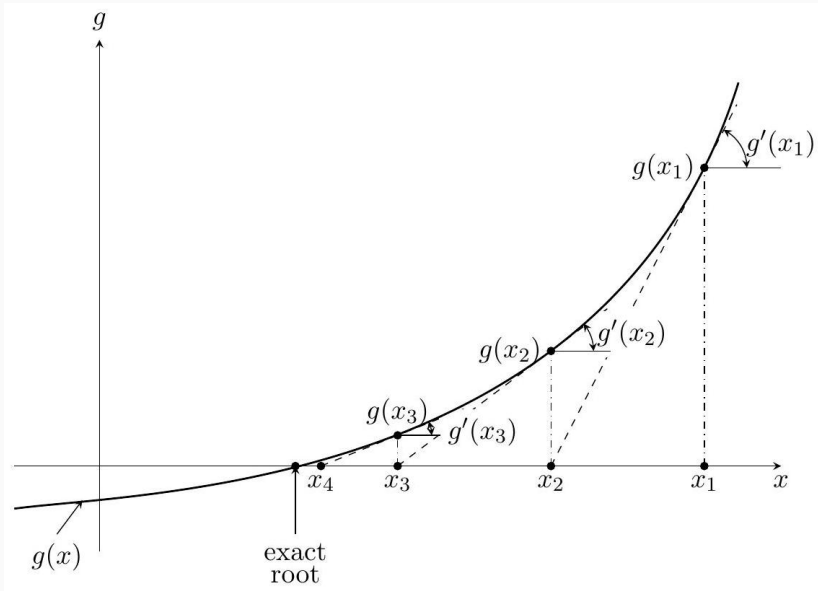


Figure 2: Newton method applied to find root of g .

If ℓ is the tangent line to f at x , then $\ell(y) = f(x) + f'(x)(y - x)$, and $\ell(y) = 0$ when

$$y = x - \frac{f(x)}{f'(x)}$$

Iterating this procedure using y as the next best guess, we obtain

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, \dots$$

Ideally, $x_n \rightarrow x_*$, where $f(x_*) = 0$. Newton's method can fail, for example, if f' is unbounded or is zero. Thus we need to identify a class of functions or impose conditions for which we can find zeros.

Example

Suppose we want to find the square root of a number r (e.g. $r = 17$).

We can frame this as a root-finding problem: find x such that $g(x) = x^2 - r = 0$.

Newton's method for this problem is:

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)} = x_n - \frac{x_n^2 - r}{2x_n} = \frac{1}{2} \left(x_n + \frac{r}{x_n} \right).$$

Let's try it with $r = 17$ and $x_0 = 4$:

$$x_1 = \frac{1}{2} \left(4 + \frac{17}{4} \right) = 4.125$$

If we repeat this process, we get the following:

n	x_n
0	<u>4.0000000000000000</u>
1	<u>4.1250000000000000</u>
2	<u>4.123106060606061</u>
3	<u>4.123105625617683</u>
4	<u>4.123105625617661</u>

Now the actual square root of 17 is approximately 4.123105625617661.

The number of correct digits doubles every iteration, a phenomenon known as **quadratic convergence**.

Here we provide a rough idea of why Newton's method converges quadratically - in the next lecture and in the homework we will look at rigorous versions of such convergence proof.

Assume $f \in C^2$, and $x_0 \in (x_* - \eta, x_* + \eta)$ for some $\eta > 0$ to be determined later.

Let $\varepsilon_n = x_n - x_*$ be the error at n th step. Then

$$\varepsilon_{n+1} = x_{n+1} - x_* = x_n - \frac{f(x_n)}{f'(x_n)} - x_* = \varepsilon_n - \frac{f(x_n)}{f'(x_n)} = \frac{\varepsilon_n f'(x_n) - f(x_n)}{f'(x_n)}.$$

By Taylor's Theorem,

$$0 = f(x_*) = f(x_n - \varepsilon_n) = f(x_n) - \varepsilon_n f'(x_n) + \frac{1}{2} \varepsilon_n^2 f''(\xi_n)$$

for some ξ_n between x_n and x_* . Rearranging,

$$\varepsilon_n f'(x_n) - f(x_n) = \frac{1}{2} f''(\xi_n) \varepsilon_n^2$$

Thus

$$\varepsilon_{n+1} = \frac{1}{2} \frac{f''(\xi_n)}{f'(x_n)} \varepsilon_n^2 \leq c(\eta) \varepsilon_n^2$$

where

$$c(\eta) = \frac{1}{2} \frac{\max |f''(x)|}{\min |f'(x)|}$$

where the maximum and minimum run over all x such that $|x - x_*| \leq \eta$.

In fact,

$$\lim_{\eta \rightarrow 0} c(\eta) = \frac{1}{2} \frac{f''(x_*)}{f'(x_*)}$$

since $f \in C^2$. So we can pick η small enough such that

$$\rho = \eta c(\eta) < 1$$

Since $x_0 \in (x_* - \eta, x_* + \eta)$ we have $|\varepsilon_0| = |x_0 - x_*| \leq \eta$ and $|\xi_0 - x_0| \leq \eta$. By definition of $c(\eta)$,

$$\frac{1}{2} \left| \frac{f''(\xi_0)}{f'(x_0)} \right| \leq c(\eta)$$

It follows that

$$\begin{aligned} |\varepsilon_1| &= |x_1 - x_*| \leq c(\eta) \cdot \varepsilon_0^2 \\ &= |\varepsilon_0| \cdot |\varepsilon_0| \cdot c(\eta) \\ &\leq |e_0| \cdot \eta \cdot c(\eta) \\ &= \rho |\varepsilon_0| \end{aligned}$$

Since $\rho \leq 1$, it follows that x_1 is also in $(x_* - \eta, x_*, \eta)$ since. Repeat argument to get

$$|\varepsilon_2| \leq \rho |\varepsilon_1| \leq \rho^2 |\varepsilon_0|.$$

By induction, we obtain

$$|\varepsilon_n| \leq \rho^n |\varepsilon_0|.$$

Since $0 < \rho < 1$, the error term converges to 0 . That is, Newton's method converges, and

$$\varepsilon_{n+1} \leq c(\eta) \varepsilon_n^2$$

so the sequence converges quadratically.